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Q.T. 1

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Queuing Theory

* Concept of Queuing theory

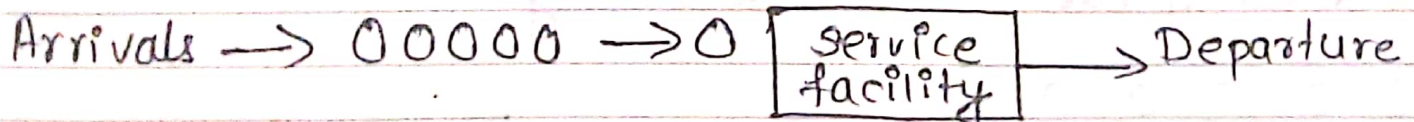
The line or list of customers for services in service sectors is known as queue. The queue may be temporary or permanent.

The analysis of waiting line of customers, behaviour of customers, pattern of their arrival and their departure, service time, etc is called the queuing theory. The main objective of queuing theory is to minimize the waiting time in minimum operating cost.

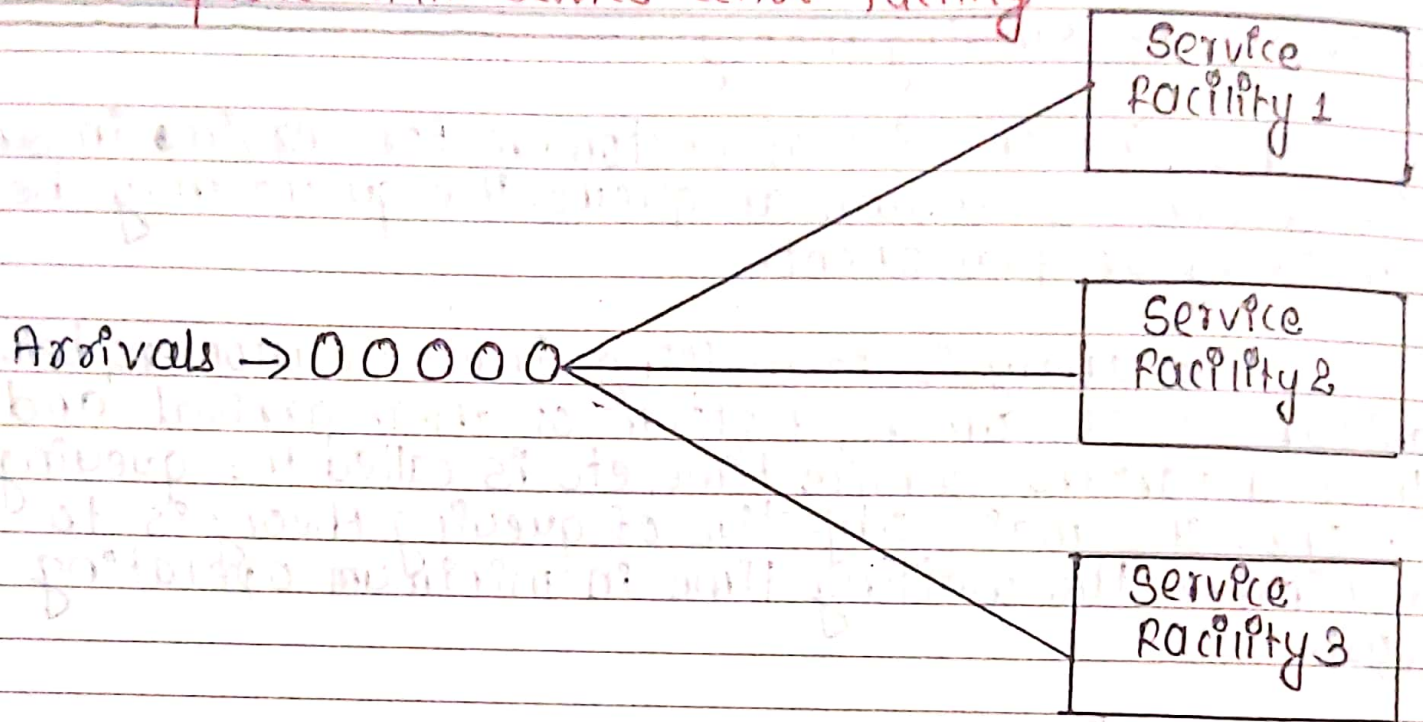
* Service Process/Queuing Models

The manner in which customers are served and leave the system is called the service process. There are mainly five types of service process:-

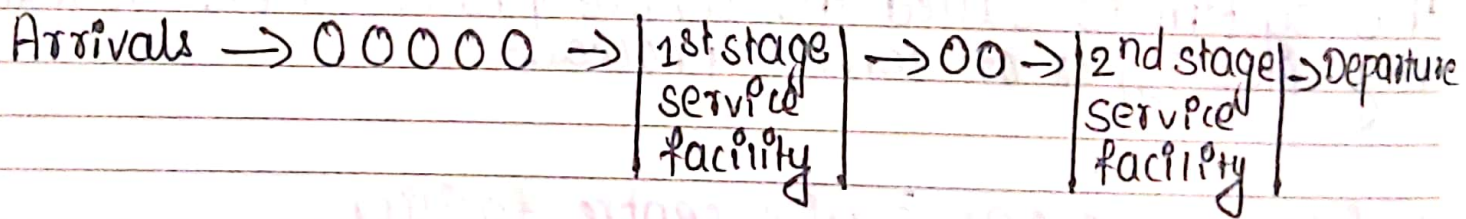
1. One queue one service centre facility



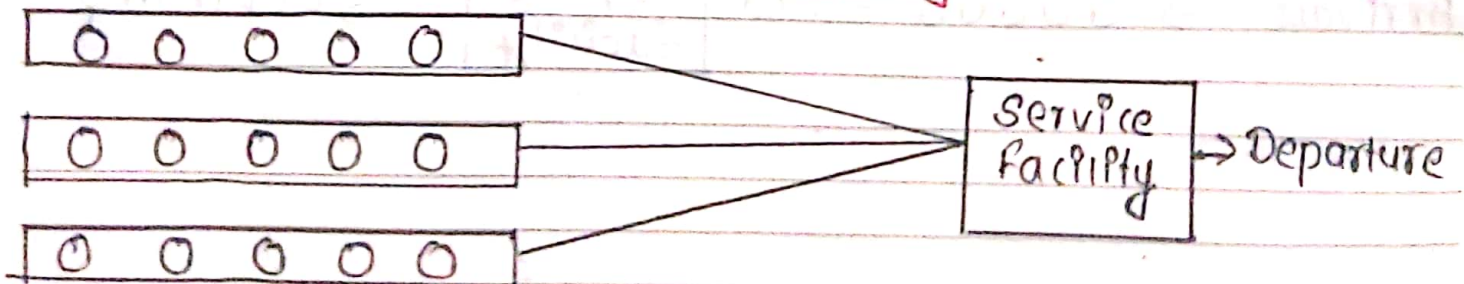
2. One queue multi-service centre facility



3. Single queue multi stage facility



4. Several queue one service facility



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Q.T.3

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5. Multi-queues and multi-service facilities

| 0 0 0 0 0 | → | service facility 1 | →

| 0 0 0 0 0 | → | service facility 2 | →

| 0 0 0 0 0 | → | service facility 3 | →

* Operating characteristics of Queuing Theory

There are five operating characteristics of queuing theory:-

1. Queue length

The average number of customers waiting in the queue for making service is known as queue length. It is denoted by L_q and given by,

$$L_q = \frac{\lambda}{\mu} \left(\frac{\lambda}{\mu - \lambda} \right)$$

2. System length

The average number of customers in the system (sum of them waiting in the queue and one customer is getting service) is known as system length. It is denoted by L_s and given by,

$$L_s = \frac{\lambda}{\mu - \lambda}$$

3. Waiting time in queue

The average time spent in the queue for waiting service before starting his service is known as waiting time in queue. It is denoted by T_q and given by,

$$T_q = \frac{\lambda}{\mu} \left(\frac{1}{\mu - \lambda} \right)$$

4. Waiting time in system

The average time spent by a customer in the system before completing the service is known as waiting time in system. It is denoted by T_s and given by,

$$T_s = \frac{1}{\mu - \lambda}$$

5. Utilization Factors / Traffic Intensity

The percentage or proportion of time that the system is busy is known as utilization factors or traffic intensity. It is denoted by ρ and given by,

$$\rho = \frac{\lambda}{\mu}$$



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Q.T.5

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* Formulae in Queuing theory

(i) Probability that the system is free $(P_0) = 1 - P$

(ii) Probability that there are k customers in the system,

$$P(n=k) = \left(1 - \frac{\lambda}{\mu}\right) \left(\frac{\lambda}{\mu}\right)^k$$

(iii) Probability that there are at least k customers in the system,

$$P(n \geq k) = \left(\frac{\lambda}{\mu}\right)^k$$

(iv) Probability that there are more than k customers in the system,

$$P(n > k) = \left(\frac{\lambda}{\mu}\right)^{k+1}$$

where,

λ = Arrival Rate

μ = Service Rate